The Information Order of Experiments: Intervention Lattices and Generator Identifiability

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Abstract

Which experiment should you run next? We give the question a finite, exactly computable form. For a hypothesis space $\mathcal{H}$ of candidate generators, write $O_1 \preceq O_2$ when O_1 is obtainable from O_2 by garbling. Refinement then contracts preimages, $P_{O_2}(g) \subseteq P_{O_1}(g)$, so per-generator ambiguity $i_O(g)$ and its population mean $H(G \mid O)$ both fall—a one-line consequence, but one that turns "more informative" into an order rather than a scale.

We instantiate the order exactly on all 1,500,625 four-state partially observed Markov chains, enumerating every subset of hidden-state resets. Four findings. The passive experiment is a *deterministic* coarsening of the full interventional one, confirmed exactly, so its 1.874 bits of shortfall is precisely $I(G; O_I \mid O_P)$. Informativeness is genuinely partial, not total: a single reset leaves 4.504 bits where passive observation leaves 3.245, so one intervention is *worse* than watching, while two are better. Which two matters enormously and the rule is structural—two resets inside one observational lump identify 0% of the class, two that cross the lump identify 58.55%, at identical cost. And information gain is submodular across every triple we can form, which places greedy experiment selection on familiar footing.

The organising claim is that resolving a mechanism is partition refinement, that the order on experiments controls its rate, and that the interventions worth buying are the ones that cross distinctions the observation map erases.

1 Introduction

A companion paper [5] argues that identifiability should be measured rather than assumed, by counting the preimage $P_O(g) = \{h \in \mathcal{H} : O(h) = O(g)\}$ of an observation map, and reports that identifiability and predictive uncertainty are distinct coordinates. It leaves a question open. Identifiability there is relative to a declared observation map and a declared class of admissible interventions, and changing that class moved 82% of one hypothesis space between regimes. If knowability is regime-relative, the regimes themselves need a structure.

They have one. Experiments are partially ordered by informativeness, and the order is old: Blackwell's comparison of experiments [1] ranks one experiment above another when the second can be obtained from the first by a stochastic garbling. What we add is a finite, exactly enumerable instance in which the whole lattice can be computed rather than argued about, together with three empirical facts that are not consequences of the order alone.

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2 The refinement order

Let $\mathcal{H}$ be finite, G uniform on $\mathcal{H}$, and let O_1, O_2 be observation maps.

Definition 1. *$O_1 \preceq O_2$ iff there is a map F with $O_1 = F \circ O_2$.*

Proposition 2 (Containment)**.** *If $O_1 \preceq O_2$ then for every $g \in \mathcal{H}$, $P_{O_2}(g) \subseteq P_{O_1}(g)$, hence $i_{O_2}(g) \leq i_{O_1}(g)$ and $H(G \mid O_2) \leq H(G \mid O_1)$.*

Proof. If $O_2(h) = O_2(g)$ then $O_1(h) = F(O_2(h)) = F(O_2(g)) = O_1(g)$. Containment gives the inequality on $\log_2$ of cardinalities pointwise, and averaging preserves it. □

Proposition 3 (Value of refinement)**.** *If $O_1 \preceq O_2$ then $H(G \mid O_1) - H(G \mid O_2) = I(G; O_2 \mid O_1)$.*

Proposition 3 is what licenses reading a difference of conditional entropies as the information the finer experiment carries that the coarser one cannot reach. It requires the order; between incomparable experiments the difference is a comparison of two numbers and nothing more.

Proposition 4 (One-way migration)**.** *Classify generators by thresholds on $i_O(g)$ and on a behavioural coordinate $p_k(g)$. If p_k is evaluated under an operating condition held fixed across $O_1 \preceq O_2$, then under refinement no generator can become less identifiable, and none can move in the behavioural coordinate.*

This is worth stating because it bounds what a migration statistic can mean. Reported migration under refinement is necessarily one-dimensional and one-directional; it is not evidence of two-dimensional mobility.

3 The benchmark system

We reuse the family of [5, §9] rather than introduce another: hidden state $X_t \in \{0, 1, 2, 3\}$, transition rows drawn from $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ summing to one (35 rows), all $35^4 = 1{,}500{,}625$ generators enumerated completely, observation lumping $O(0) = O(1) = 0$, $O(2) = O(3) = 1$, horizon $T = 6$. The passive experiment starts from the uniform hidden state. For $A \subseteq \{0, 1, 2, 3\}$, the experiment O_A resets the hidden state to each $j \in A$ in turn and returns the resulting observable laws.

4 Results

The passive experiment is a deterministic coarsening. Writing L_j for the law obtained by resetting to state j, we verified exactly, over every generator, that

$$O_P \;=\; \tfrac{1}{4} \textstyle\sum_j L_j,$$

so $O_P \preceq O_{\{0,1,2,3\}}$ with a deterministic F. Proposition 2 then applies, and we confirm its conclusion computationally as an implementation check: zero containment violations across every pair $A \subset B$, and zero against the passive mixture.

Informativeness is partial, not total. A single reset leaves 4.504 bits; passive observation leaves 3.245. One intervention is *less* identifying than watching a system you did not touch. There is no contradiction with Proposition 2: O_P is a coarsening of the full reset set, not of any singleton, so the two are incomparable in $\preceq$ and the proposition says nothing about them. Two resets (2.618) overtake passive; one does not. Any claim of the form "intervention beats observation" is therefore false as stated, and the correct statement is that the order is partial.

Reset set A	$H(G \mid O_A)$	classes	$\Pr[i_{O_A} = 0]$	$\max \lvert P \rvert$
$\varnothing$	20.517	1	0.00%	1,500,625
any singleton	4.504	400,584	0.00%	67,375
$\{0,1\}$ or $\{2,3\}$	3.356	482,902	0.00%	30,625
$\{0,2\},\{0,3\},\{1,2\},\{1,3\}$	2.618	967,107	58.55%	8,953
any triple	1.914	1,101,899	69.84%	7,263
$\{0,1,2,3\}$	1.371	1,235,809	82.01%	6,561
passive (mixture)	3.245	303,229	0.00%	7,537

Table 1: The intervention lattice, exact over 1,500,625 generators. Rows within a group are identical to the digits shown, reflecting the symmetry of the construction.

The interventions that pay are those crossing the observation's kernel. Let $x \sim_O y$ iff $O(x) = O(y)$, here $\{0,1\}$ and $\{2,3\}$. Two resets chosen inside one $\sim_O$ class leave 3.356 bits and identify *nothing*; two that cross leave 2.618 bits and uniquely identify 58.55% of the entire hypothesis class. The cost is identical. Greedy selection, run without knowledge of the lumping, chooses 0, then 2, then 3, then 1—crossing at the first opportunity.

The mechanism is not mysterious: resets within a $\sim_O$ class differ only in a distinction the observation map already erases, so they re-ask a question the passive experiment has largely answered. We state it as a conjecture rather than a theorem, since we have measured it on one lumping of one family:

> *Crossing conjecture.* Interventions that separate states identified by O are more informative about G than interventions of equal cardinality confined within a single $\sim_O$ class.

Information gain is submodular. With $F(A) = H(G \mid O_\varnothing) - H(G \mid O_A)$, every one of the 108 triples $A \subseteq B$, $j \notin B$ satisfies $F(A \cup \{j\}) - F(A) \geq F(B \cup \{j\}) - F(B)$, with no violations. Submodular maximisation under a cardinality constraint admits the classical $(1 - 1/e)$ greedy guarantee [4], so where the property holds, greedy intervention selection is not merely a heuristic. The greedy gain sequence here is 16.013, 1.886, 0.704, 0.543 bits: the fourth reset is nearly redundant.

The value of full intervention. $H(G \mid O_P) - H(G \mid O_I) = 3.245 - 1.371 = 1.874$ bits, which by Proposition 3 is exactly $I(G; O_I \mid O_P)$: the generator information the complete reset protocol carries that no amount of passive observation can reach.

5 Related work

The order is Blackwell's [1]; our instance is its degenerate deterministic case. Choosing experiments to maximise expected information is Lindley's [3], and choosing interventions to secure identifiability is well developed in causal discovery, where optimal strategies are known [2]. The greedy guarantee for submodular objectives is [4]. What we contribute is not a new criterion but an exactly enumerated lattice: every experiment in a finite family, every containment relation checked rather than assumed, and the resulting empirical facts—incomparability, the crossing rule, submodularity—measured rather than argued.

6 Limitations

The lattice is computed on one family with one lumping, and the striking symmetry of Table 1 is a property of that construction. The crossing conjecture is supported by a single 2+2 lumping of four states and should not be treated as established. Submodularity is confirmed on 108 triples in one family and is not proved; we do not know what conditions on $\mathcal{H}$ and O imply it, and that is the obvious next theorem. Identification is at horizon $T = 6$, which is where the partition of this family stabilises [5, §9], but stability over a finite window is evidence rather than proof. Proposition 2 is a theorem, so the computation confirming it tests our implementation, not the mathematics.

7 Discussion

Resolving a mechanism is partition refinement. Beginning with $[g] = \mathcal{H}$, each experiment induces an equivalence relation and the cell containing the truth contracts,

$$\mathcal{H} \supseteq [g]_{\Pi_1} \supseteq [g]_{\Pi_2} \supseteq \cdots,$$

with perfect resolution at $[g]_\Pi = \{g\}$ and irreducible ambiguity when the admissible lattice bottoms out while $|[g]_\Pi| > 1$. What the measurements add is that the rate of contraction is controlled by the order on experiments, that the order is partial rather than total—so more control is not uniformly more information—and that the experiments worth buying are the ones that cross the distinctions the observation map erases.

Reproducibility. `mgs_lattice.py` produces every figure here; `audit.py` re-derives each against it. Both are in the repository accompanying [5].

References

[1] D. Blackwell. Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2):265–272, 1953.

[2] A. Hauser and P. Bühlmann. Two optimal strategies for active learning of causal models from interventional data. *International Journal of Approximate Reasoning*, 55(4):926–939, 2014.

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